

Week 2

Data Types and Formats

ISTA 322 - Data Engineering

This week in data engineering

- Based on Slack intros we have some pretty diverse skill sets
 - Some with lots of coding but maybe less python
 - Some with more DS data skills
 - Some with a bit of both
 - This is a good thing!
- This week: quick review of different ways data are represented via data types and in different formats
 - Might not be new for some of you - that's ok
- Also start working with data (2 Jupyter notebooks + HW1 due Sept 5th)

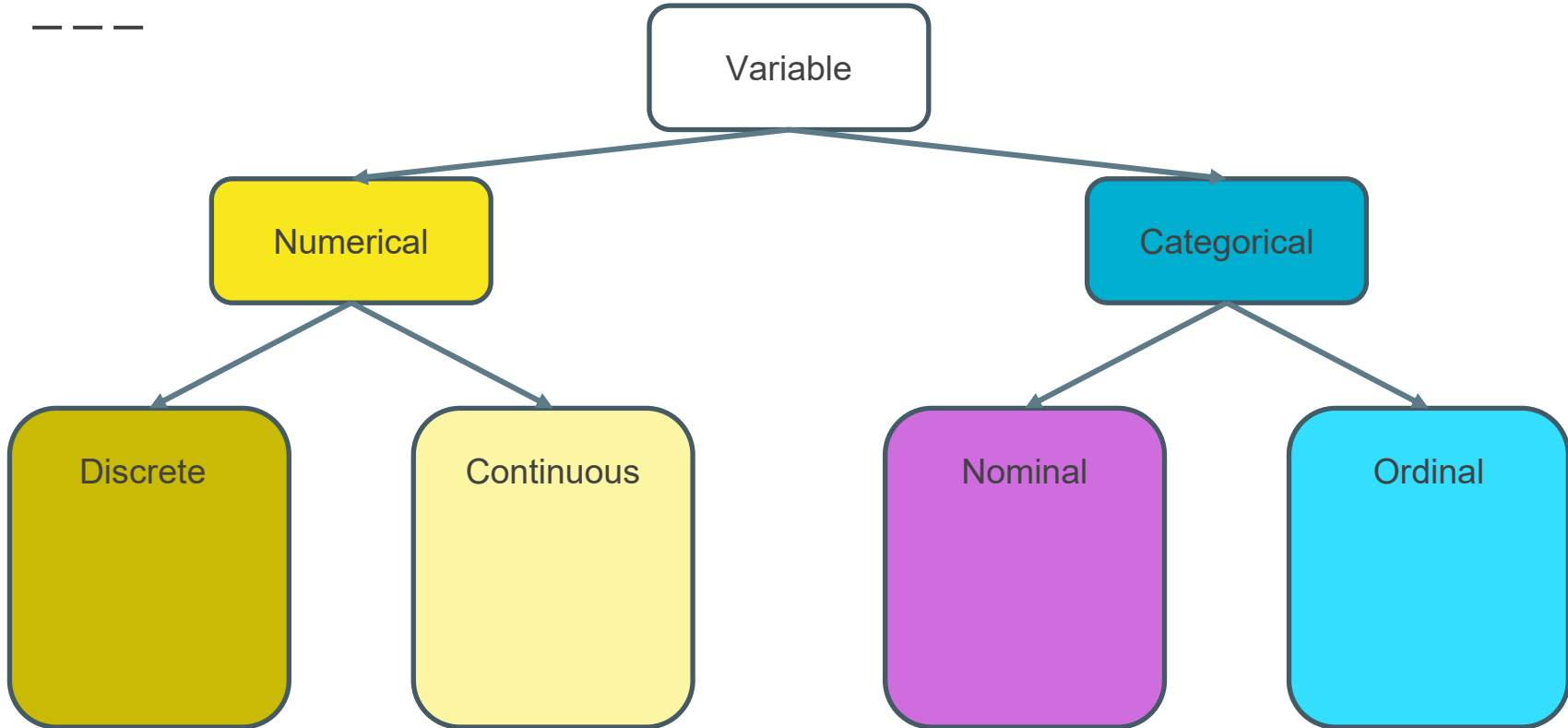
What is data anyway?

- **Data** - Collection of facts consisting of numbers/words that are taken through measurements with the goal of describing things
- Let's create data on me
 - Get weight - 230lbs
 - Get height - 72"
 - Hair color - brown
- These are measurements that describe me
 - Level of precision is determined by tool used
 - Also at data entry
- Obviously numbers and words

Describing data

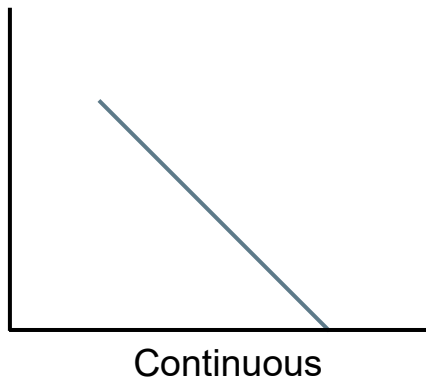
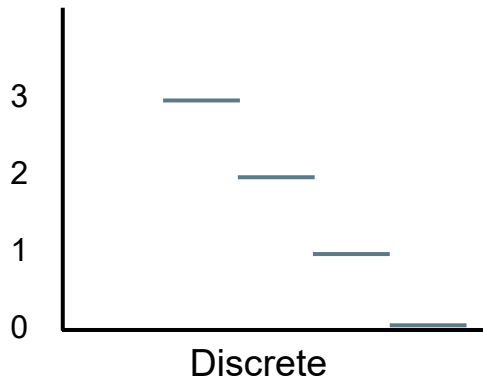
- Should be simple: numbers and words
- But there's more nuance than that
- Numbers
 - Continuous vs discrete, binary vs non-binary
- Words
 - Nominal vs ordinal, numbers stored as text, text coded as numbers, T/F
- Nuance is important to accurately describe reality
- Practical reasons too (e.g. storage, modeling)

Describing data - Overview



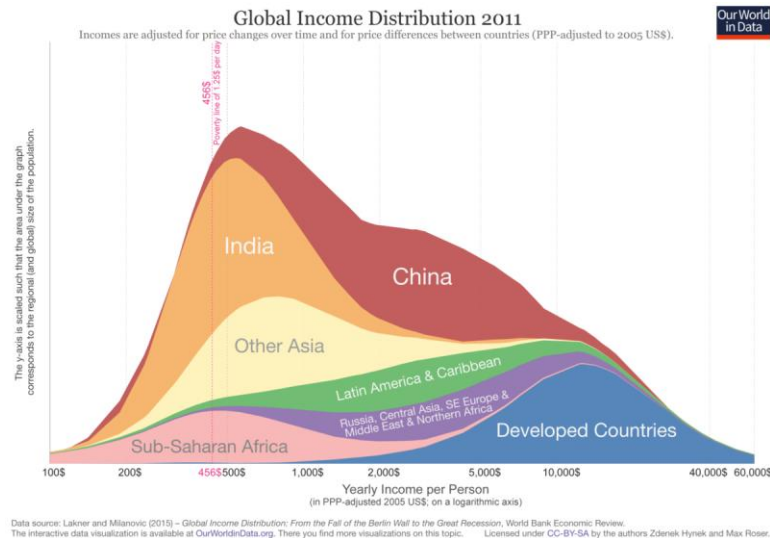
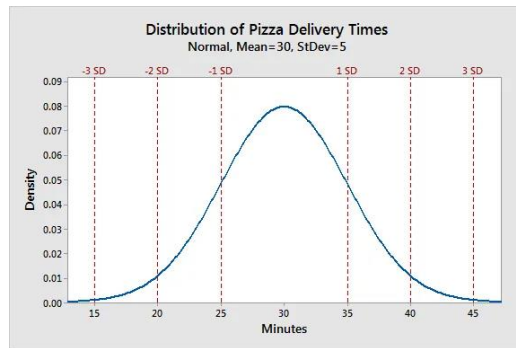
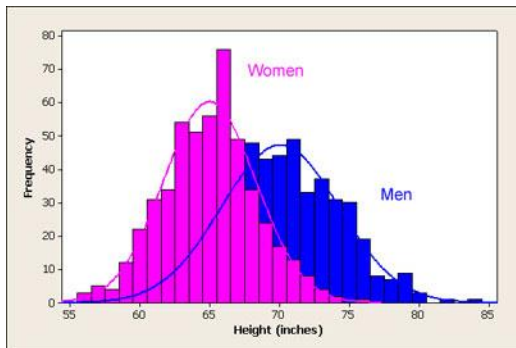
Continuous vs. discrete

- Useful to think whether data is continuous or discrete
- Discrete - data that can only have certain values
 - Limited amount of 'in-between' numbers
- Continuous - data that can take any value (continuum)
 - Any 'in-between' number is possible



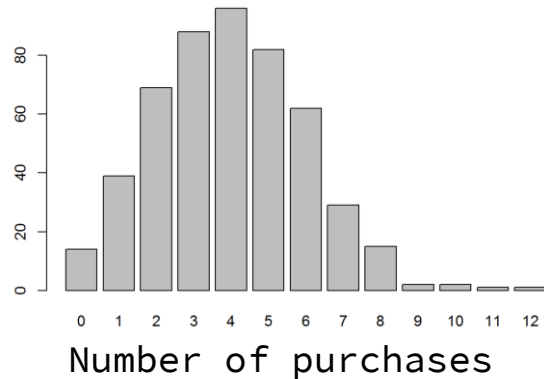
Datatypes - Numeric continuous - Float

- Continuous numeric datatypes you're familiar with
- Height, weight, sales per store, etc
 - Can have any value in a range
 - Have a decimal
- Float datatype



Datatypes - Numeric discrete - Integer

- Number of clicks, number of sales, passengers on a plane
- These are numeric
 - Can have any whole value in a range
- But they're not continuous
 - Can't have fractional values
 - No half clicks, half sales, half passengers, etc
- Integer datatype



Datatypes - Numeric discrete - Binary

- Binary datatypes are very common to indicate yes/no or present/absent
- yes = 1, no = 0

purchased	purch_b
yes	1
no	0
no	0
no	0
yes	0
no	0

Datatypes - Numeric discrete - Binary

- Binary datatypes are very common to indicate yes/no or present/absent
- yes = 1, no = 0
- Many ML models need numeric
- Can encode levels of strings
 - One hot encoding
- Integer datatype

OS			
windows			
android			
mac			
mac			
windows			
android			

Datatypes - Numeric discrete - Binary

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OS	win_b	and_b	mac_b
windows	1	0	0
android	0	1	0
mac	0	0	1
mac	0	0	1
windows	1	0	0
android	0	1	0

Nominal vs. ordinal

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- Ordinal
 - Values have a hierarchy/relative value to each other
- Nominal
 - Values are independent of each other



Which of the following items do you normally choose for your pizza toppings? (select all that apply)

☐ Spinach

☒ Pepperoni

☐ Olives

☒ Sardines

☐ Sausage

☐ Extra cheese

☐ Onions

☒ Tomatoes

☐ Other (please specify):

To what extent do you agree with the following statement: The company made it easy for me to handle my issue.

- ☒ Strongly agree
- ☐ Agree
- ☐ Partly agree
- ☐ Disagree
- ☐ Strongly disagree

x

What is your annual income?

- ☐ Less than \$15,000
- ☐ \$15,000 to \$34,999
- ☐ \$35,000 to \$49,999
- ☐ \$50,000 to \$74,999
- ☐ \$75,000 to \$99,999
- ☐ \$100,000 or more

Datatypes – Categorical - Strings

- Strings are any length of alphanumeric characters
- Could be nominal – aka text!
 - 'black', 'brown', 'blonde', 'red'
 - 'Pineapple on pizza is good, change my mind'
 - 'Y'all shouldn't be having parties with 100 people'
 - '2013-01-12 07:54:22 __RequestToken_Lw__=2B3CC Channie Tununak'
- Could be ordinal
 - 'small', 'medium', 'large'
 - 'extremely unhappy', 'unhappy', 'neutral', 'happy', 'extremely happy'
- String datatype

Datatypes - Categorical - Boolean

- Booleans are a lot like binary
 - Two values, True and False
 - Not strings, though
- True = 1
- False = 0
- Frequently used in logical statements
 - If x is TRUE, then do y
- Boolean datatype

Datatypes - Datetimes

- Time data is really important for rolling up data
 - Sales in a hour/day/month
 - Average monthly temperature
 - Number of requests per minute
- Frequently imported as strings
 - '02/18/2020 12:12:47'
- But can't do critical operations with them as strings
- Casting to datetime allows program to understand these as continuous values
- Datetime datatype
 - Not native in python - need 'datetime' module
 - SQL does have them

Datatypes

- The exact syntax and methods you'll use to manipulate these will vary across tools
- And there's more nuance within
- But it's important to think about the data type and if the values make sense for its type
- Again, probably review for many of you
 - But still can't be said enough :)

Moving on to Data Structures

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- Data structures are ‘how’ data types are organized
 - Numeric data stored in a list: [67, 49, 88, 95, 77]
 - Numeric data is a data type
 - List is the way it’s stored
- Lists, data frames, and dictionaries
- Most have seen, but we’ll run through them quick

Lists

- Ordered sequence of elements
- `Test_scores = [67, 49, 88, 95, 77]`
 - `print(test_scores)`
 - Output: `[67, 49, 88, 95, 77]`
- Can be updated, added to, sliced, etc using different methods in python
- Can also be strings or mixed
 - `Dan_info = [230, 72, 'brown']`
- `[]`

Dataframe

- Not native to python - Need Pandas
- Just a 2-D object with labeled columns and rows
- Multiple lists bound together
 - `Test_scores = [67, 49, 88, 95, 77]`
 - `Study_time = [35, 14, 75, 89, 68]`
 - `School_year = ['fr', 'jr', 'sr', 'sr', 'fr']`
- Pandas is very powerful for data wrangling

test_scores	study_time	school_year
67	35	fr
49	14	jr
88	75	sr
95	89	sr
77	68	fr

Dictionary

- Dictionaries are key-value pairs
 - {key:value , key:value}
- `Dan_dict = {name: 'dan', height: 72, weight: 230}`
- Dictionaries can be searched quickly if you know the key
- Can be converted to data frames
- JSON files are organized sets of key-value pairs

Exciting huh?

- OK, not the most thrilling topic, but we need to make sure we're all on the same page
- Other part of this week will be coding
- But just need to cover a bit on several key data formats
 - Flat files, relational database, json

Flat files

- Flat files refer to 2 dimensional data
 - csv, tsv
- Each row represents an observation
- Only one file containing all info
- Often great for analysis
 - It's a data frame!
- Slower to search
- Inefficient storage

transact_id	store_id	store_state	country	UPC	price
x88943	az_23	AZ	USA	49914	2.57
x88943	az_23	AZ	USA	99371	1.99
a85921	to_39	Ontario	Canada	95831	8.99
a85921	to_39	Ontario	Canada	99492	5.49
a85921	to_39	Ontario	Canada	27482	4.49
z88930	az_45	USA	USA	33491	0.99

Relational database

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- Goal will be to put transformed data into relational database
 - The 'L' in ETL
- Tables 'relate' to one another based on keys
- Efficient
- Can be queried in many ways (e.g. SQL)

TABLE ID: STORE		
store_id	store_state	country
az_23	AZ	USA
az_45	AZ	USA
ca_12	CA	USA
to_39	Ontario	Canada

TABLE ID: TRANSACTIONS			
transact_id	store_id	UPC	price
x88943	az_23	49914	2.57
x88943	az_23	99371	1.99
a85921	to_39	95831	8.99
a85921	to_39	99492	5.49
a85921	to_39	27482	4.49
z88930	az_45	33491	0.99

JSON

- Semi-structured
- Need to parse and load into DB
- Keys = column names
- Values = observations
- Can then query

```
{
  "contributors": null,
  "coordinates": null,
  "created_at": "Fri Jun 28 07:31:35 +0000 2019",
  "display_text_range": [
    0,
    1
  ],
  "entities": {
    "hashtags": [],
    "symbols": [],
    "urls": [
      {
        "display_url": "twitter.com/polhomeeditor/\u0266",
        "expanded_url": "https://twitter.com/polhomeeditor/status/1144289510739587073",
        "indices": [
          2,
          25
        ],
        "url": "https://t.co/0fgkUFjCaB"
      }
    ]
  },
  "user_mentions": [
    {
      "favorite_count": 3,
      "favorited": false,
      "full_text": "? https://t.co/0fgkUFjCaB",
      "geo": null,
      "id": 1144508626738044929,
      "id_str": "1144508626738044929",

```

Table: Tweet_info		
t_id	t_time	t_coord
1144..	Fri Jun	null

Table: Tweet_urls		
t_id	disp_url	url
1144..	twitter.com.	t.co/0f...

Table: Tweet_social		
t_id	n_fav	favorited
1144..	3	false

Recap

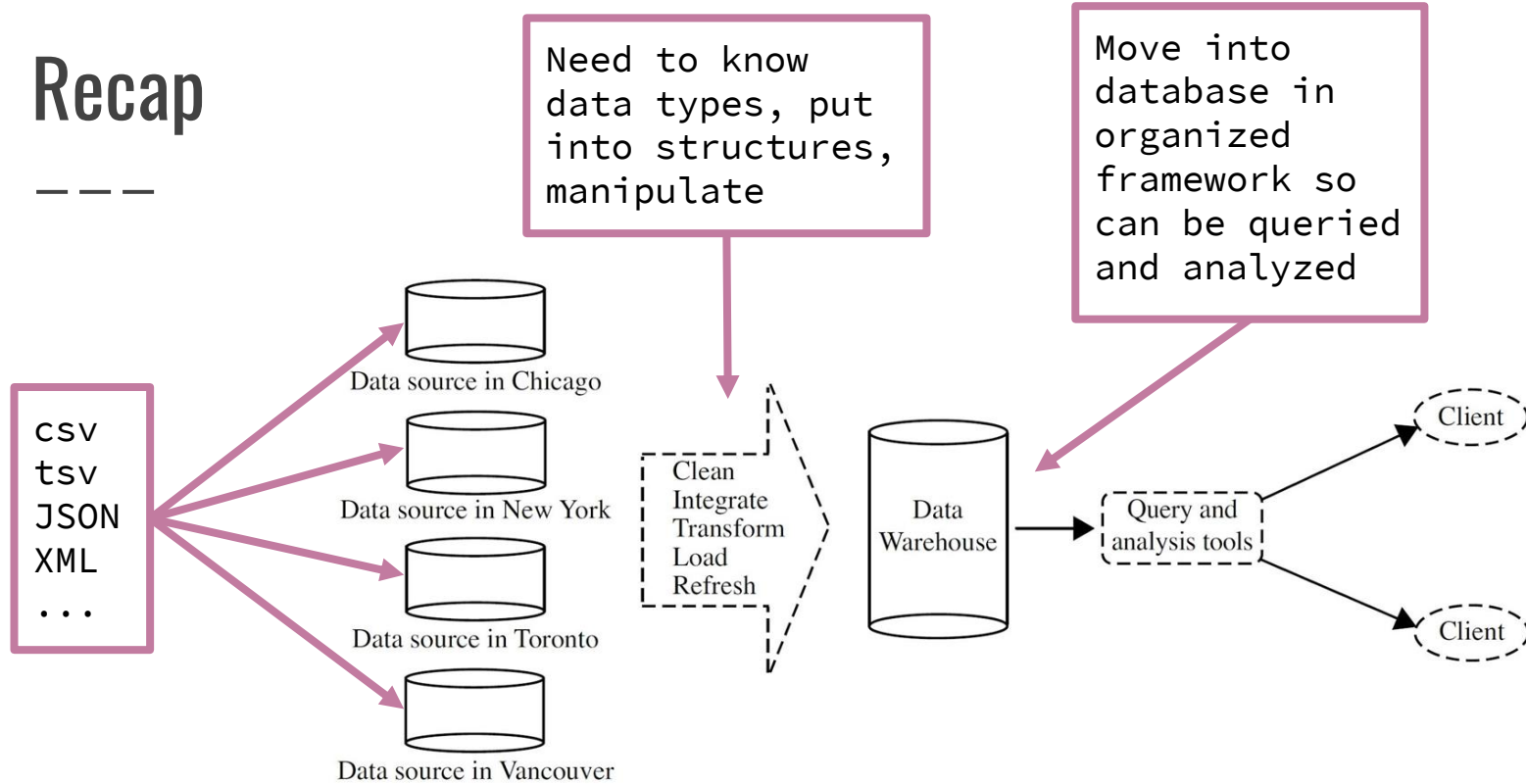


Figure I.6 Typical framework of a data warehouse for *AllElectronics*.

Recap

- It's ok if some bits are not clear
 - Just to make sure everybody is on the same general page
- These are more concepts that we will be applying across languages
- Other part of this week (and rest of the class) we'll be applying these ideas and more.

Tasks for this week

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- Jupyter notebooks
 - Google colab and Python basics
 - Functions and loops
 - Homework 1
 - * Make a copy on your google drive to work on it.
- When done, download as .ipynb and upload to D2L